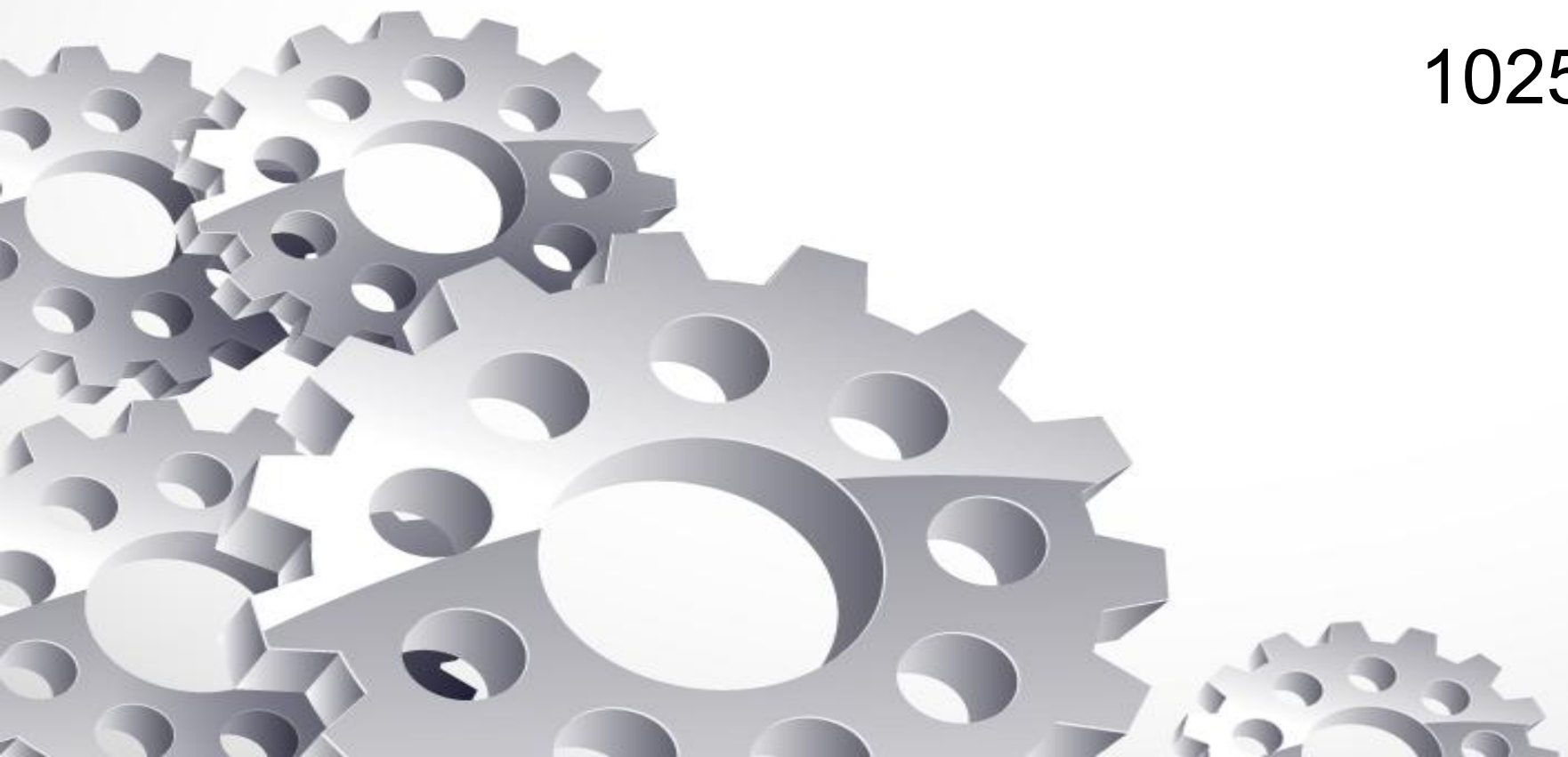


Music Classification

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Introduction



- background

Massive Expansion of Digital Music Libraries: Platforms such as QQ Music and NetEase Cloud Music host millions of songs, creating vast repositories of digital music.

Urgent Need for Automated Classification: Manual labeling of music genres is time-consuming and labor-intensive, necessitating automated solutions.

- Motivation

Limitations of Traditional Methods

Constraints of Handcrafted Feature Approaches: Methods based on manual feature extraction (e.g., MFCC combined with traditional machine learning) perform poorly in complex scenarios.

Rise of Deep Learning

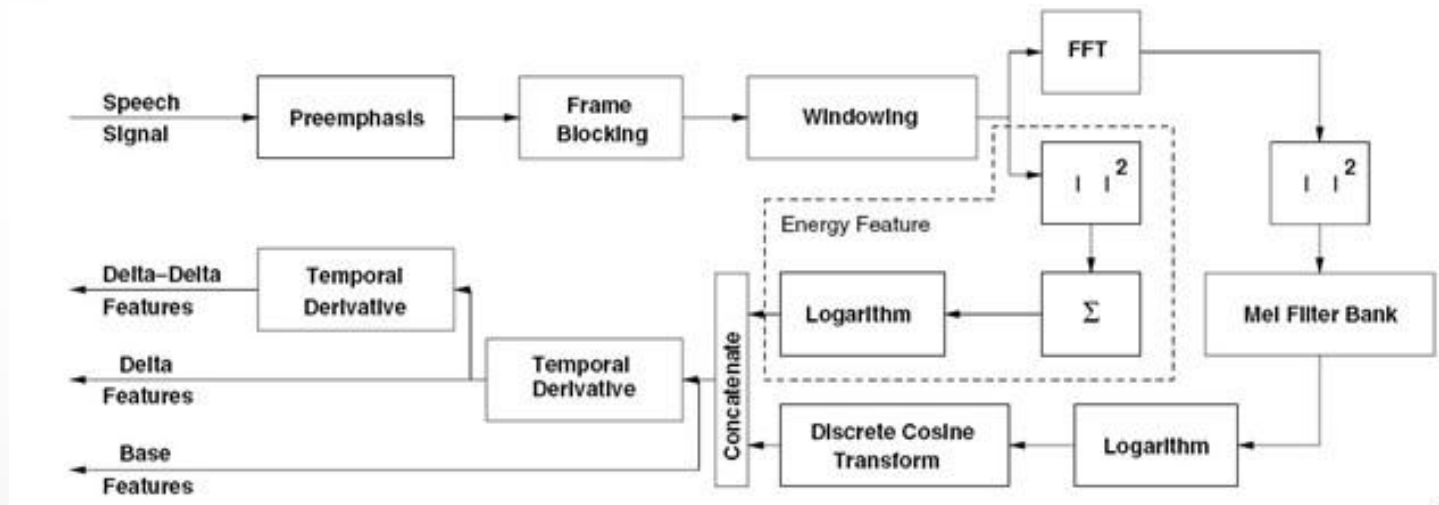
Success of Convolutional Neural Networks (CNNs): The achievements of CNNs in image classification have opened new avenues for audio classification.

Related Works

- **MFCC**

MFCC is a widely used feature extraction technique in speech recognition and audio processing. It simulates the nonlinear characteristics of the human auditory system, converting audio signals into compact feature representations.

Processing Pipeline:



Related Works

- **librosa**

Librosa is a Python toolkit specifically designed for music and audio analysis. It enables you to easily convert audio files (such as WAV, MP3) into digital signals that computers can process, and perform feature extraction and visualization.



Related Works

- **CNN**

CNN (Convolutional Neural Network) is a specialized deep learning architecture designed for processing grid-like data, such as images or audio spectrograms. It employs convolutional layers to automatically and hierarchically extract local features through filters, while significantly reducing parameters via weight sharing. This structure enables efficient learning of spatial or temporal patterns, making it foundational to modern computer vision and audio analysis.



Problem Statement



- **Current Challenges**

Complex Music Data: Audio signals are high-dimensional, time-varying complex information

Manual Classification Not Scalable: Facing massive music libraries, manual labeling is inefficient, costly, and subjective

Complex Feature-Genre Relationship: Multi-dimensional features such as rhythm, melody, harmony, and timbre require comprehensive analysis

- **Project Objectives**

Develop an end-to-end deep learning solution capable of:

Automatically extracting distinctive features from raw audio

Accurately identifying 10 common music genres

Achieving over 80% test accuracy

Providing interpretable analysis of model decisions

Data Processing

Dataset:

GTZAN Genre Collection (from kaggle)

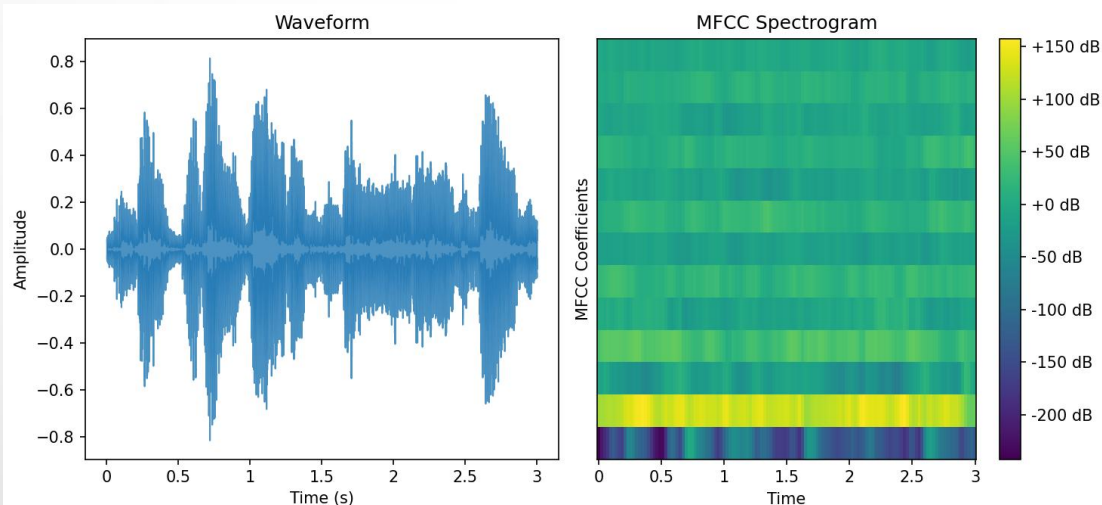
A collection of 10 genres with 100 audio files each, all having a length of 30 seconds (the famous GTZAN dataset, the MNIST of sounds)



Data Processing

Audio->Vector

Extract MFCC features using librosa, standardize the MFCC features, and then reshape the data into the desired format (batch, channels, height, width) for CNN, where the number of channels is 1, the height represents the number of time frames, and the width corresponds to the MFCC coefficients.



```
try:
```

```
# 加载音频
signal, sr = librosa.load(file_path, sr=SAMPLE_RATE)

# 切片处理 (30s / 3s = 10个片段)
num_segments = int(track_duration / segment_duration)

for d in range(num_segments):
    start = samples_per_segment * d
    finish = start + samples_per_segment

    # 提取MFCC
    mfcc = librosa.feature.mfcc(
        y=signal[start:finish],
        sr=sr,
        n_mfcc=num_mfcc,
        n_fft=n_fft,
        hop_length=hop_length
    )
    mfcc = mfcc.T # 转换为 (时间, 频率)

    # 确保所有切片的形状完全一致
    if len(mfcc) == num_mfcc_vectors_per_segment:
        # 标准化
        mfcc = (mfcc - mfcc.mean()) / (mfcc.std() + 1e-8)

        # 添加通道维度 [1, 时间, 频率]
        mfcc = torch.FloatTensor(mfcc).unsqueeze(0)

        self.data.append(mfcc)
        self.labels.append(genre)
```

Model

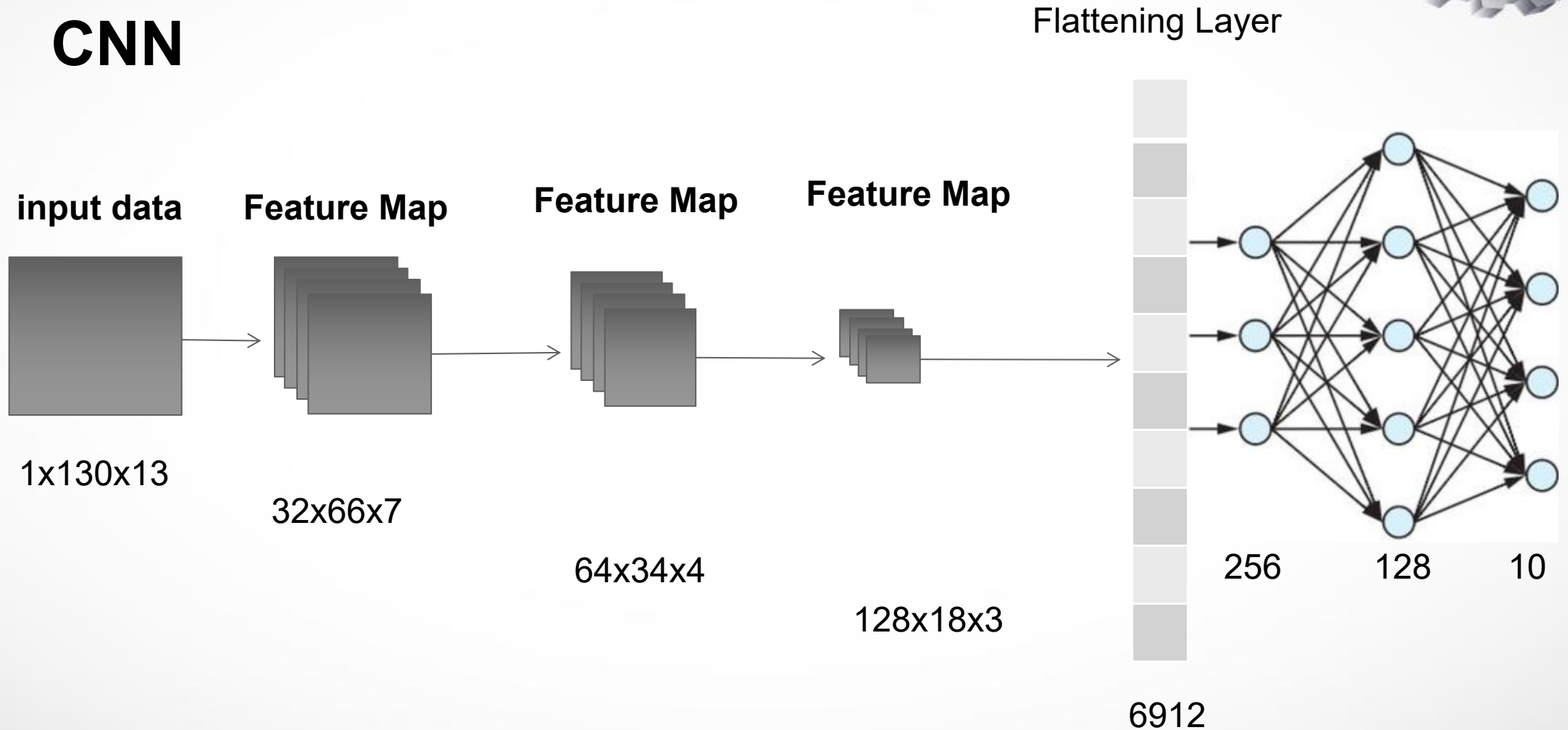


--- 模型架构可视化 ---

```
=====
Layer (type:depth-idx)           Output Shape           Param #
=====
MusicCNN                         [1, 10]                --
├─Conv2d: 1-1                    [1, 32, 130, 13]       320
├─BatchNorm2d: 1-2              [1, 32, 130, 13]       64
├─MaxPool2d: 1-3                [1, 32, 65, 7]         --
├─Conv2d: 1-4                   [1, 64, 65, 7]         18,496
├─BatchNorm2d: 1-5              [1, 64, 65, 7]         128
├─MaxPool2d: 1-6                [1, 64, 33, 4]         --
├─Conv2d: 1-7                   [1, 128, 33, 4]        73,856
├─BatchNorm2d: 1-8              [1, 128, 33, 4]        256
├─MaxPool2d: 1-9                [1, 128, 17, 3]        --
├─Linear: 1-10                  [1, 256]                1,671,424
├─Dropout: 1-11                 [1, 256]                --
├─Linear: 1-12                  [1, 128]                32,896
├─Dropout: 1-13                 [1, 128]                --
├─Linear: 1-14                  [1, 10]                 1,290
=====
```

Model

CNN



Evaluation

- Accuracy

1. 基础性能指标:

准确率 (Accuracy): 0.8070

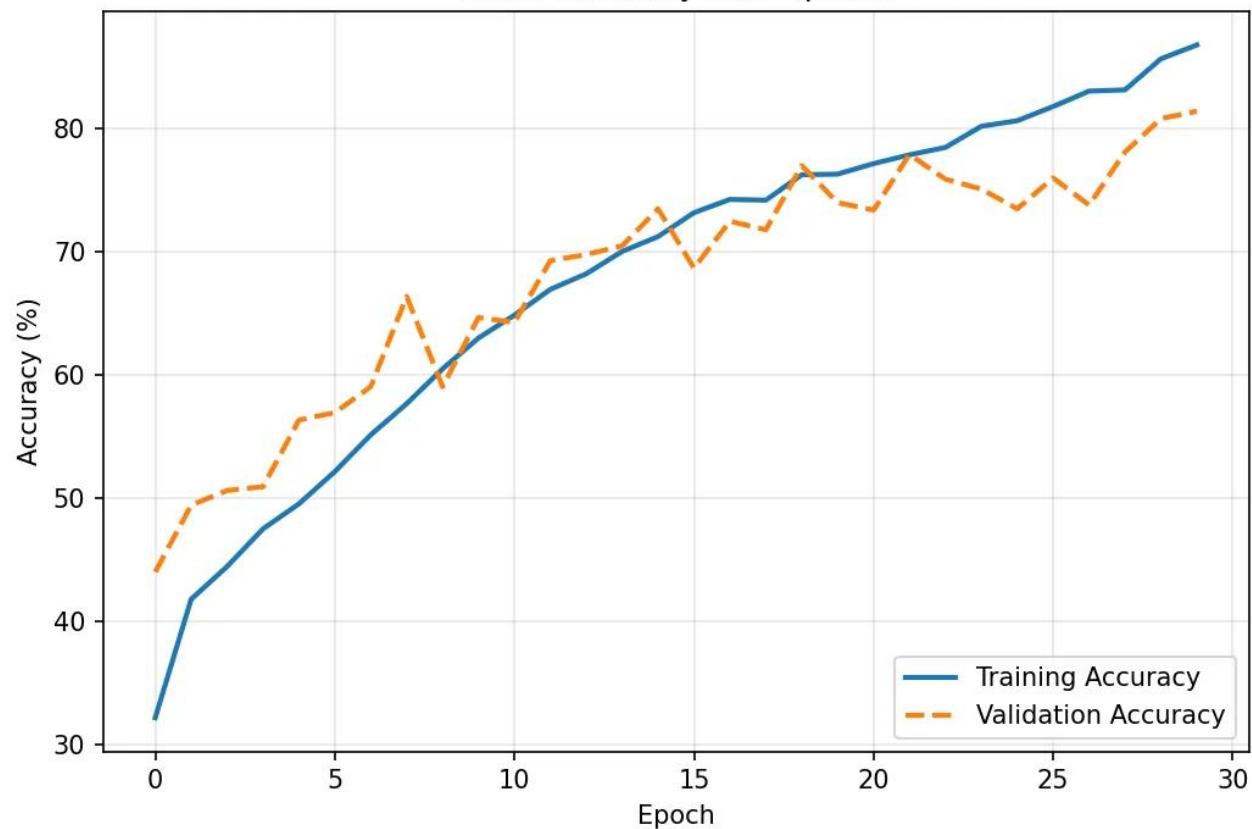
宏平均精确率 (Precision): 0.8111

宏平均召回率 (Recall): 0.8047

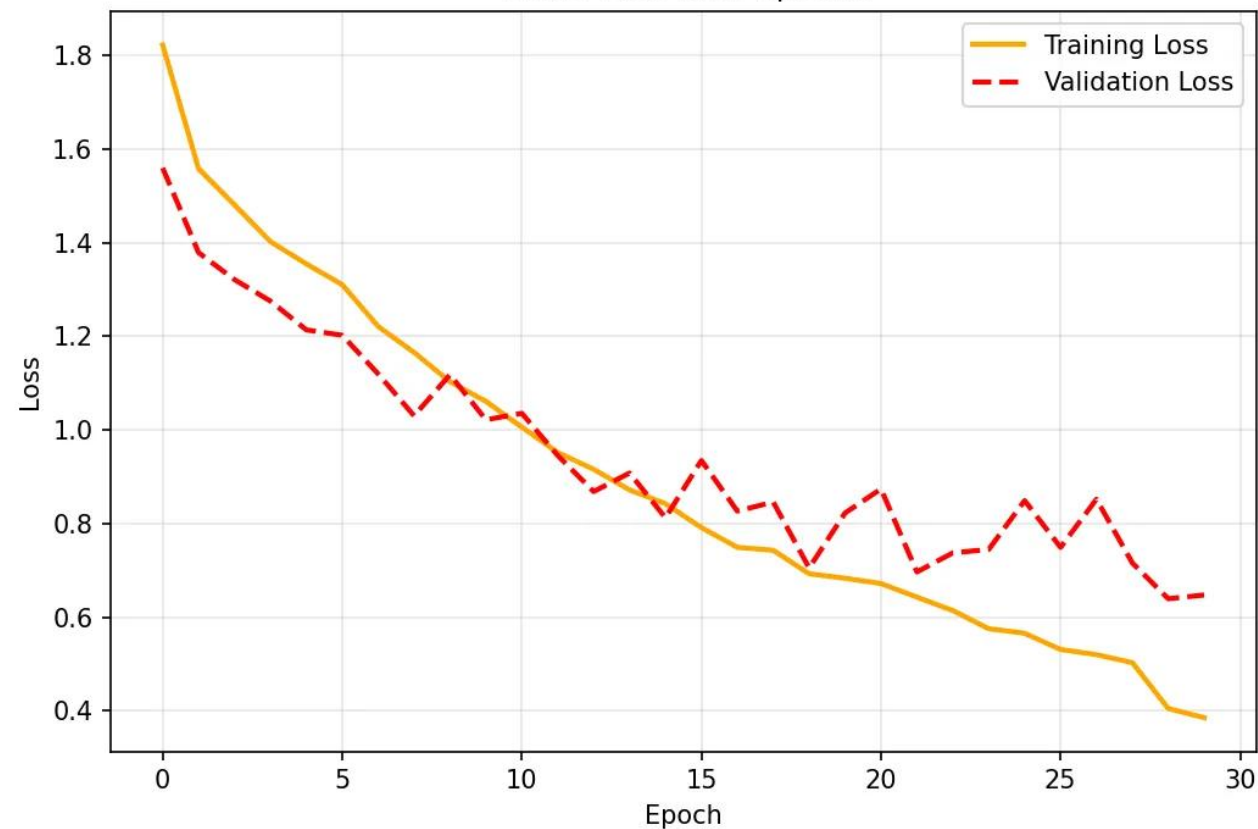
宏平均F1分数: 0.8054



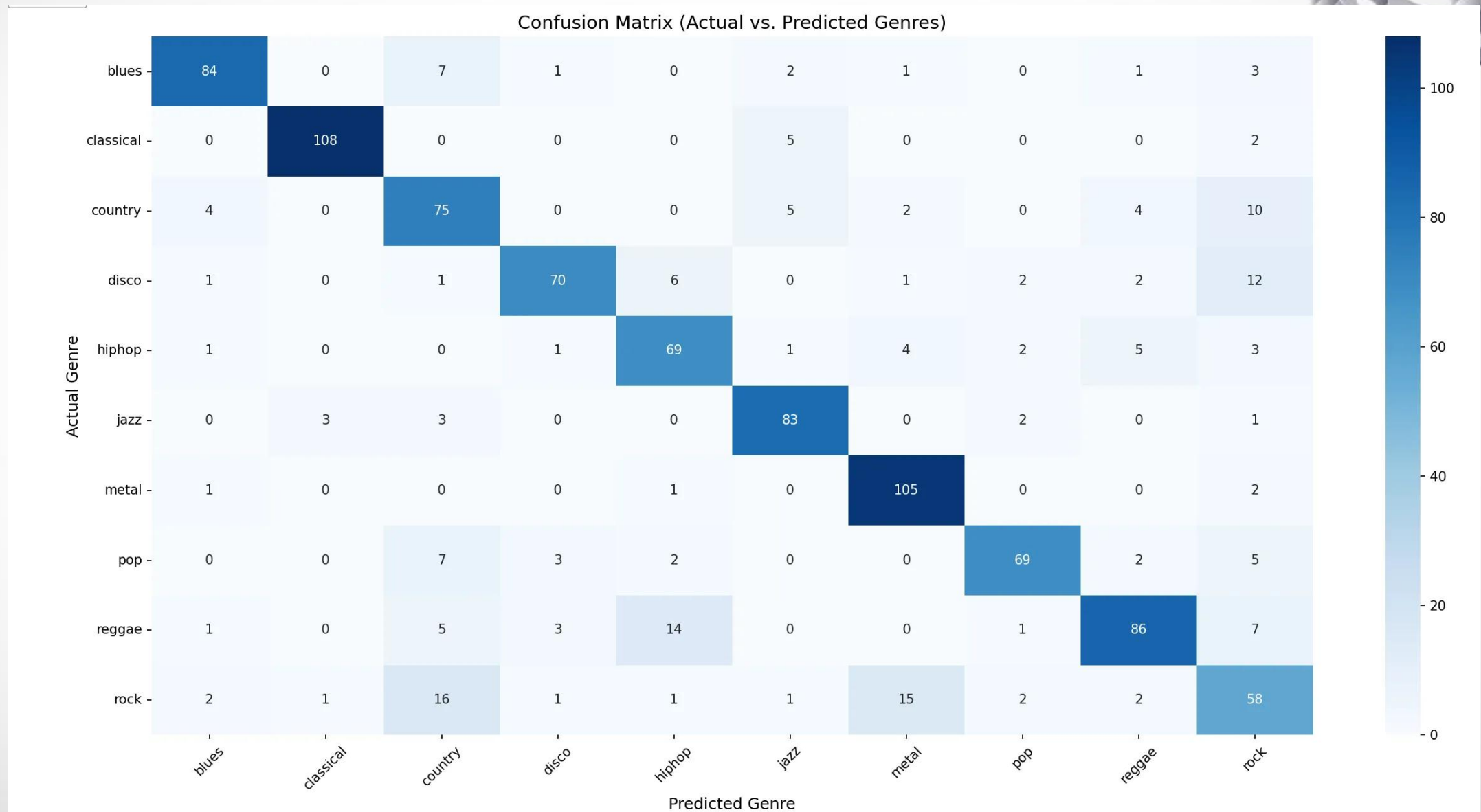
Model Accuracy over Epochs



Model Loss over Epochs



Evaluation



Conclusion

- This system has successfully demonstrated the effectiveness of deep learning in music genre classification, providing a well-structured and extensible framework. Through a meticulously designed CNN architecture and a comprehensive evaluation system, we have not only achieved high classification accuracy but also gained a deep understanding of the model's decision-making process and limitations. This solution offers a practical technical reference for the field of Music Information Retrieval (MIR), with potential for further research and commercialization.





- Thank you for listening !